Four-Corner Spectral Tuning for Quadratic ADMM with Application to Diffeomorphic Image Registration

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Abstract

Selecting the penalty and relaxation parameters of the alternating direction method of multipliers (ADMM) can require repeated spectral-radius evaluation of a large parameter-dependent iteration operator. We consider over-relaxed ADMM (oADMM) for a separable quadratic split whose attained joint modal spectrum is a Cartesian product of data and regularizer curvatures. The modal contraction factor is monotone along each curvature coordinate line; consequently, its largest magnitude is attained among four endpoint combinations. The relaxation parameter is optimized analytically for each penalty, and the remaining penalty problem reduces to a finite algebraically characterized superset of candidate minimizers. In this sense the method is search-free: it avoids grid-based or iterative numerical optimization over the penalty while still computing the required polynomial roots numerically. The assembled diffeomorphic-registration operator is spatially varying and noncommuting, so the exact theorem does not apply to it. A family of globally coefficient-frozen surrogates indexed by the observed local optical-flow blocks, together with the rank-one Hessian structure, nevertheless yields an inexpensive predictor determined by the four combinations of h_-, h_+, g_-, and g_+. On 134 FIRE retinal pairs at 256×256, one-shot tuning reduced median pairwise runtime by 34.4% and inner iterations by 39.6% relative to a validation-selected fixed comparator chosen on separate images. On ten CIMA histology pairs, runtime decreased by 20.1%. In both studies, landmark accuracy was essentially unchanged and no nonpositive discrete Jacobian was observed.

Keywords: alternating direction method of multipliers; parameter selection; spectral analysis; diffeomorphic image registration; local Fourier analysis

1 Introduction

Spectral parameter selection for quadratic ADMM can be expensive in its own right. For a trial penalty ρ, a spectral method constructs or applies the corresponding iteration operator, estimates its convergence factor, and repeats this calculation while optimizing ρ; over-relaxation introduces the additional parameter α. In large imaging systems the operator may have hundreds of thousands of unknowns, so repeated spectral-radius evaluation can offset the computational advantage of splitting. This issue is especially pronounced after spatially varying image terms are assembled and whenever relinearization changes the local Hessian.

The structural question addressed here is whether this spectral optimization can collapse analytically when the quadratic operators possess separable modal structure. For the class studied below, the oADMM contraction of a common mode depends only on a data curvature h and a regularizer curvature g. For fixed ρ, this scalar factor is monotone along each coordinate line,

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with the direction determined by the other curvature. If the attained joint modal spectrum is the Cartesian product of the two curvature sets, the worst contraction must therefore occur among

$$(h_-, g_-), \quad (h_-, g_+), \quad (h_+, g_-), \quad (h_+, g_+).$$

The large spectral optimization is reduced exactly to four rational scalar branches. The optimal relaxation is available in closed form for each penalty, and the reduced penalty objective is piecewise rational. Its global minimum on a prescribed compact interval belongs to a finite algebraically characterized candidate set.

We call this parameter selection *search-free*: it performs neither a grid search nor an iterative numerical optimization over ρ. Polynomial roots in the finite candidate construction are still evaluated numerically; the distinction is the replacement of repeated optimization and spectral-radius evaluation by a fixed algebraic reduction.

The exact result requires more than commutativity. Simultaneous diagonalization supplies paired eigenvalues (h_i, g_i), but these pairs need not fill the Cartesian product of the marginal spectra. The four endpoint combinations are exact modes only under the separability and attainment assumptions stated in Theorem 4.2. For arbitrary commuting operators they define a rectangular envelope, and for noncommuting operators they define a surrogate unless an independent bound is available.

Diffeomorphic image registration provides the motivating noncommuting application. Its linearized data Hessian varies by pixel, whereas the Sobolev regularizer couples neighboring pixels; the assembled matrices therefore generally do not commute. Freezing the image coefficient recovers the separable model through a family of globally constant surrogates indexed by the observed local Hessian blocks. Moreover, each optical-flow Hessian is a rank-one update of a multiple of the identity, so all required data curvature information reduces to

$$h_- = \nu, \qquad h_+ = \nu + \mu \max_i \|\nabla I(x_i)\|^2 .$$

Together with the regularizer endpoints g_- and g_+, these values yield a four-branch predictor after one $O(n)$ scan of gradient magnitudes. The result is exact for the frozen separable family and predictive for the full variable-coefficient registration operator.

The mathematical and computational progression is

$$\begin{aligned}\text{quadratic ADMM} &\longrightarrow \text{modal factor} \longrightarrow \text{four corners}\\ &\longrightarrow \text{finite parameter selection} \longrightarrow \text{noncommuting registration.}\end{aligned}$$

The contributions are:

1. A four-corner spectral theorem for quadratic oADMM under an attained Cartesian-product joint modal spectrum.

2. A finite algebraic joint selection of ρ and α that eliminates grid-based or iterative penalty search for this structured model.

3. A registration specialization based on local coefficient freezing and the rank-one optical-flow Hessian, reducing the data-term information required for tuning to endpoint curvatures.

4. Experimental evidence that the surrogate-optimal parameters transfer to noncommuting registration systems, reducing iterations and runtime while maintaining landmark accuracy and observed topology.

2 Related Work

2.1 ADMM parameter selection

Scaled ADMM and its practical residual criteria were systematized by Boyd et al. [3]. Ghadimi et al. derived optimal parameters for specific quadratic classes [5]. Teixeira et al. connected parameter selection with scaling and constraint preconditioning in distributed quadratic programs [13], while Giselsson and Boyd developed tight contraction bounds that depend on the chosen metric [6]. These works obtain strong conclusions from particular quadratic structure or operator bounds. The present analysis instead identifies a joint modal structure under which the full worst-mode calculation reduces to four attained endpoint pairs.

Adaptive methods avoid an offline spectral model. Residual balancing changes the penalty according to primal and dual residual magnitudes [17]; Barzilai–Borwein (BB) spectral updates estimate local curvature from iterate differences [18]. Such rules apply beyond the separable setting considered here, but they update parameters from the observed iteration history rather than solving the fixed quadratic model's spectral minimax problem algebraically.

2.2 Spectral and over-relaxed ADMM

Song et al. address a broader class of linear-quadratic ADMM problems and numerically optimize the spectral convergence factor with respect to the penalty [12]. For each candidate ρ, their formulation requires spectral information from the corresponding iteration operator; the relaxation parameter is then selected conditionally. This generality is valuable because it does not require the Cartesian-product modal structure used here.

Direct application to full-resolution registration is computationally demanding: the assembled variable-coefficient operator is large and noncommuting, and its spectrum changes when image warping changes the linearized Hessian. Repeated spectral-radius evaluation across trial penalties can therefore make parameter selection expensive. A downsampled operator reduces this cost but introduces a transfer problem between the coarse spectral model and the full-resolution solve. The present work adopts the complementary tradeoff. Stronger separability assumptions collapse the structured spectral problem to four scalar branches and a finite penalty candidate set; the resulting construction is then used as an inexpensive predictor outside that exact class. A coarse direct-spectral diagnostic illustrating this overhead and transfer issue is reported in Supplementary Section 11.2.

2.3 Fourier and local Fourier analysis

Fourier analysis has long characterized stationary iterations for elliptic problems [4], and local Fourier analysis (LFA) is a standard tool for multigrid design and parameter optimization [16]. Translation-invariant periodic problems admit exact Fourier decompositions; local and nonstandard analyses can remain informative for random or discontinuous coefficients [11, 9]. Here the stationary map is an ADMM error iteration, and global coefficient freezing converts each observed optical-flow block into a separable constant-coefficient surrogate.

2.4 Variational image registration

DARTEL [1], diffeomorphic demons [15], and large-deformation optimal-control formulations [10] are representative optimization-based registration methods. Thorley et al. compose nonstationary velocity updates and solve regularized convex subproblems with accelerated ADMM on graphics processing units (GPUs) [14]. Song et al. use a related quadratic model in their spectral oADMM study [12]. Learning-based methods broaden the current landscape [7], but the present work concerns numerical parameter selection for the linearized variational subproblem.

3 Quadratic ADMM and Reduced Error Dynamics

3.1 Quadratic splitting

Consider

$$\min_{v,w} \underbrace{\tfrac{1}{2}v^\top Hv + b^\top v}_{f(v)} + \underbrace{\tfrac{1}{2}w^\top Gw}_{r(w)} \quad \text{subject to} \quad v - w = 0, \tag{1}$$

where $v, w, b \in \mathbb{R}^N$ and $H, G \in \mathbb{R}^{N\times N}$ are symmetric positive semidefinite matrices. The linear term b determines the fixed point but disappears from homogeneous error propagation.

3.2 Standard and over-relaxed ADMM

With scaled dual variable u and penalty $\rho > 0$, standard ADMM is

$$(H + \rho\mathrm{I})v^{k+1} = -b + \rho(w^k - u^k), \tag{2}$$

$$(G + \rho\mathrm{I})w^{k+1} = \rho(v^{k+1} + u^k), \tag{3}$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \tag{4}$$

For over-relaxed ADMM, abbreviated *oADMM*, define

$$\widehat{v}^{k+1} = \alpha v^{k+1} + (1-\alpha)w^k, \qquad 0 < \alpha \le 2, \tag{5}$$

and replace v^{k+1} by $\widehat{v}^{k+1}$ in (3) and the dual update.

3.3 Reduced Cayley representation

The original oADMM recurrence couples primal and dual errors. Reducing it to one N-dimensional operator isolates all dependence on ρ and α and exposes the common-mode scalar factor used in the four-corner argument. Define

$$C_H(\rho) = (H - \rho\mathrm{I})(H + \rho\mathrm{I})^{-1}, \qquad C_G(\rho) = (G - \rho\mathrm{I})(G + \rho\mathrm{I})^{-1}.$$

Proposition 3.1 (Reduced Cayley error operator)**.** *The nonzero spectra of the $2N$-dimensional oADMM state map for (w, u) and*

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right)\mathrm{I} + \frac{\alpha}{2}C_H(\rho)C_G(\rho). \tag{6}$$

coincide, including algebraic multiplicity; the remaining state-map eigenvalues are zero. Thus the asymptotic factor is $\varrho(E_{\rho,\alpha})$, where $\varrho(\,\cdot\,)$ denotes the spectral radius of a matrix or linear operator.

Appendix A gives the reduction. It is supporting machinery: the parameter reduction below comes from the modal dependence of (6).

Theorem 3.2 (Basic contraction)**.** *If $H, G \succeq 0$ and $H + G \succ 0$, then for every $\rho > 0$, $\kappa_\rho := \|C_H(\rho)C_G(\rho)\|_2 < 1$. Consequently*

$$\varrho(E_{\rho,\alpha}) \le \|E_{\rho,\alpha}\|_2 \le 1 - \frac{\alpha}{2}(1 - \kappa_\rho) < 1,$$

for $0 < \alpha \le 2$.

Proof. For $A \succeq 0$, orthogonal diagonalization shows $\|C_A(\rho)\|_2 \le 1$, with norm preservation possible only on $\ker A$. If a unit vector x satisfied $\|C_H C_G x\|_2 = 1$, equality throughout $\|C_H C_G x\|_2 \le \|C_G x\|_2 \le 1$ would give $x \in \ker G$ and $C_G x = -x \in \ker H$. Hence $x \in \ker H \cap \ker G$, contradicting $H + G \succ 0$. The continuous function $x \mapsto \|C_H C_G x\|_2$ attains its maximum on the finite-dimensional unit sphere, so $\kappa_\rho < 1$. Finally, the triangle inequality applied to (6) gives the stated bound. □

Nullspaces. If a nonzero vector lies in $\ker H \cap \ker G$, then $C_H C_G x = x$ and $E_{\rho,\alpha} x = x$ for every admissible parameter pair. Strict contraction therefore requires $H + G \succ 0$, or restriction to the identifiable subspace. In the four-corner registration model, $\nu = \gamma = 0$ creates the $(h, g) = (0, 0)$ mode, for which $|e(0, 0; \rho, \alpha)| = 1$ identically. Positive damping or a coercive regularizer endpoint removes that mode from the modal product. If instead the common null mode is quotiented out explicitly, the four-corner objective must be formed from the remaining attained endpoint pairs after that removal.

3.4 Scalar modal factor

When a vector is a common eigenvector of H and G, with eigenvalues h and g, (6) acts by

$$\begin{aligned} e(h, g; \rho, \alpha) &= 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h - \rho}{h + \rho} \frac{g - \rho}{g + \rho} \qquad (7) \\ &= 1 - \alpha + \alpha\theta(h, g; \rho), \\ \theta(h, g; \rho) &= \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}. \qquad (8) \end{aligned}$$

For nonnegative h and g, $0 \le \theta \le 1$. The paired values (h, g), not the marginal spectra separately, determine the true modal factor.

4 Four-Corner Spectral Tuning

4.1 Separable commuting modal structure

Definition 4.1 (Cartesian-product joint modal spectrum)**.** A quadratic split is *separable* on modal sets $\mathcal{H}$ and $\mathcal{G}$ if H and G are simultaneously diagonalizable and their attained joint modal spectrum is

$$\{(h, g) : Hx = hx,\ Gx = gx \text{ for a common mode } x \ne 0\} = \mathcal{H} \times \mathcal{G}.$$

For example, $H_{\text{sep}} = I_m \otimes H_c$ and $G_{\text{sep}} = G_s \otimes I_d$ have modes $\phi_\xi \otimes z_j$ and pairs $(h_j, g(\xi))$, where $H_c z_j = h_j z_j$, $G_s \phi_\xi = g(\xi)\phi_\xi$, $H_c \in \mathbb{R}^{d\times d}$ acts on components, $G_s \in \mathbb{R}^{m\times m}$ acts on spatial degrees of freedom, and I_m, I_d are the corresponding identity matrices.

The Cartesian-product condition is stronger than commutativity. It asserts that data and regularizer modes combine independently.

4.2 Four-corner theorem

For fixed $\rho > 0$,

$$\frac{\partial \theta}{\partial h} = \frac{\rho(g-\rho)}{(g+\rho)(h+\rho)^2}, \qquad \frac{\partial \theta}{\partial g} = \frac{\rho(h-\rho)}{(h+\rho)(g+\rho)^2}. \tag{9}$$

The sign of the first derivative is independent of h, and the sign of the second is independent of g. Thus, for fixed ρ, θ is monotone along each coordinate line, although the direction may depend on the other coordinate. Its minimum and maximum are therefore attained at corners. Relaxation maps θ affinely to $1-\alpha+\alpha\theta$; taking absolute value produces a convex function of θ, whose maximum over an interval is again attained at an endpoint.

Theorem 4.2 (Four-Corner Spectral Theorem)**.** *Suppose the joint modal spectrum is the Cartesian product $\mathcal{H} \times \mathcal{G}$, where $\mathcal{H} \subset [h_-, h_+]$ and $\mathcal{G} \subset [g_-, g_+]$, with $0 \le h_- \le h_+$ and $0 \le g_- \le g_+$. If all four endpoint combinations are attained, then*

$$\varrho(E_{\rho,\alpha}) = U_{4\mathrm{C}}(\rho, \alpha) := \max_{\substack{h \in \{h_-, h_+\} \\ g \in \{g_-, g_+\}}} |e(h, g; \rho, \alpha)| \tag{10}$$

for every $\rho > 0$ and $0 < \alpha \le 2$. More generally, if H and G are simultaneously diagonalizable and all their attained joint pairs lie in the stated rectangle, the right-hand side is a valid upper envelope even when the four corners are not attained; equality with the true spectral radius then need not hold. No such conclusion follows from the marginal intervals alone in the noncommuting case.

Proof. Equation (9) makes the extrema of θ on the rectangle occur at corners. The map $t \mapsto |1-\alpha+\alpha t|$ is convex, so its maximum over the interval between those extrema also occurs at an endpoint and hence at a corner. Cartesian-product structure and attained endpoint combinations make those corners actual modes, proving equality. $\square$

4.3 Optimal relaxation for fixed penalty

Let the four corner values satisfy

$$a(\rho) = \min_c \theta_c(\rho), \qquad b(\rho) = \max_c \theta_c(\rho).$$

The four-corner objective is

$$\max\{|1-\alpha+\alpha a|,\ |1-\alpha+\alpha b|\}.$$

When $b < 1$, equioscillation of the two endpoint factors gives the unique unconstrained optimizer; accounting for $0 < \alpha \le 2$ yields

$$\alpha^\star(\rho) = \min\left\{2, \frac{2}{2-a(\rho)-b(\rho)}\right\}. \tag{11}$$

After eliminating α, the reduced objective is

$$\begin{aligned} \widehat{U}(\rho) &= \frac{b-a}{2-a-b} && \text{if } a+b \le 1, \\ \widehat{U}(\rho) &= 2b-1 && \text{if } a+b \ge 1. \end{aligned} \tag{12}$$

If $b = 1$, the objective equals one for every admissible α; any relaxation may be selected, but the corresponding corner mode or envelope is not contractive. Appendix C gives the constrained equioscillation argument.

4.4 Finite algebraic penalty selection

After eliminating α, the objective is piecewise rational in ρ. Its expression can change only when the corner attaining $a(\rho)$ changes, the corner attaining $b(\rho)$ changes, or the relaxation regime crosses $a(\rho) + b(\rho) = 1$. Between these events, the reduced objective is one smooth rational function. A minimum on each closed piece must therefore occur at a piece endpoint or a stationary point. This converts the continuous penalty problem into finitely many algebraic conditions.

For implementation, no branch-activity test is required. Deduplicate the corner branches and index the remaining functions as $\theta_1, \ldots, \theta_m$, where $m \leq 4$. Algebraic equations are generated for every candidate branch pair; their roots form a finite superset that is filtered by evaluating the original objective.

Theorem 4.3 (Finite penalty candidate set)**.** *Let $\mathcal{I} = [\rho_-, \rho_+] \subset (0, \infty)$. A global minimizer of the reduced objective $\widehat{U}$ over $\mathcal{I}$ is contained in the following constructively generated finite set:*

1. *ρ_- and ρ_+;*
2. *roots in $\mathcal{I}$ of $\theta_i(\rho) - \theta_j(\rho) = 0$ for every distinct pair $i < j$;*
3. *roots in $\mathcal{I}$ of $\theta_i(\rho) + \theta_j(\rho) = 1$ for every pair $1 \leq i, j \leq m$; and*
4. *roots in $\mathcal{I}$ of the cleared stationary equations for*

$$F_{ij}^{(1)}(\rho) = \frac{\theta_j(\rho) - \theta_i(\rho)}{2 - \theta_i(\rho) - \theta_j(\rho)} \quad \text{and} \quad F_j^{(2)}(\rho) = 2\theta_j(\rho) - 1$$

for every pair $1 \leq i, j \leq m$.

Only positive real roots are retained. Identically zero polynomials are discarded, repeated roots are merged, and roots outside $\mathcal{I}$ are removed. After denominators are cleared, the intersection, transition, and stationarity polynomial degrees are at most three, four, and six, respectively. Evaluating the original four-corner objective at every retained candidate—using (11), or any admissible α when $b = 1$—returns a global minimizer over $\mathcal{I} \times (0, 2]$.

The proof is in Appendix D. The listed roots form a finite *superset*: equations formed from branches that are not active on a given interval and denominator clearing may introduce unnecessary candidates, but these are harmless because the original objective is evaluated afterward. Search-free selection refers to this finite algebraic evaluation in place of grid-based or iterative optimization over ρ; numerical polynomial root finding is still required.

4.5 Arbitrary commuting operators versus Cartesian products

Three cases must be distinguished.

Case A: For a separable commuting split with attained endpoint combinations, (10) is the true spectral radius, and finite tuning is globally optimal for that model over $\mathcal{I} \times (0, 2]$.

Case B: If $HG = GH$ but the paired spectrum is only $\{(h_i, g_i)\}_i$, the exact factor is $\max_i |e(h_i, g_i; \rho, \alpha)|$. The four corners of the marginal spectral rectangle can be artificial and provide an envelope rather than an equality.

Case C: If $HG \neq GH$, no global scalar modal decomposition follows. Four-corner tuning is a surrogate or predictor unless a separate argument supplies a bound.

5 Spatially Varying and Noncommuting Problems

5.1 Why global modal analysis fails

Let $H = \mathrm{diag}(H_1, \ldots, H_n)$ vary by location and let $G = G_0 \otimes I_d$ couple neighboring sites. In block form,

$$[H, G]_{ij} = (G_0)_{ij}(H_i - H_j),$$

so $HG \neq GH$ whenever coupled sites have different data blocks. The exact factor is then

$$R_{\text{true}}(\rho, \alpha) = \varrho(E_{\rho,\alpha}),$$

not a maximum of the scalar factors in (7).

5.2 Coefficient-frozen separable surrogates

For a symmetric positive semidefinite block $B \in \mathbb{R}^{d \times d}$, define the globally coefficient-frozen surrogate

$$\widetilde{H}^{(B)} = I_n \otimes B, \qquad \widetilde{G} = G_0 \otimes I_d = G.$$

The same block B is used at every spatial location; this construction is not a one-pixel spatial problem. The tensor factors commute, so the surrogate is simultaneously diagonalizable. If $Bz_j = h_j z_j$ and $G_0 \phi_\xi = g(\xi)\phi_\xi$, its common modes are $\phi_\xi \otimes z_j$, with exact modal factors $e(h_j, g(\xi); \rho, \alpha)$.

5.3 Minimax tuning over a frozen family

Let $\{B_r\}_{r=1}^m$ be a finite family of observed or representative local blocks, and let $\{h_{r,j}\}_{j=1}^d$ be the eigenvalues of B_r. If the common regularizer spectrum attains g_- and g_+, define

$$\mathcal{C}_{\text{frozen}} = \bigcup_{r=1}^{m} \bigcup_{j=1}^{d} \{(h_{r,j}, g_-), (h_{r,j}, g_+)\}.$$

One parameter pair is selected for the entire family through

$$U_{\text{frozen}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}_{\text{frozen}}} |e(h, g; \rho, \alpha)|. \tag{13}$$

Because each member of the family is a separable constant-coefficient model, (13) is the exact worst contraction factor over the entire frozen family. The finite algebraic argument of Theorem 4.3 applies unchanged to this finite collection of scalar branches.

5.4 Optional variable-coefficient rate bounds

Rigorous noncommuting bounds may be evaluated after tuning. For example, for any reference block B, let $\delta_H = \left\| H - \widetilde{H}^{(B)} \right\|_2$ and let $U_B(\rho, \alpha)$ denote the exact modal factor of that globally frozen surrogate. The Cayley resolvent identity gives

$$R_{\text{true}}(\rho, \alpha) \leq U_B(\rho, \alpha) + \frac{\alpha}{\rho}\delta_H. \tag{14}$$

The bound requires no commutativity of the assembled operators but is conservative and is not used for parameter selection. Its derivation and the block-Gershgorin, signed-rectangle, Perron-comparison, and truncated-kernel constructions are given in Supplementary Sections 2–7.

6 Application to Diffeomorphic Image Registration

6.1 Linearized registration model

Let n grid points lie in dimension $d \in \{2,3\}$, let Δx be the grid spacing, and let $q_i = \nabla I(x_i)$. At the current linearization, $v_i \in \mathbb{R}^d$ is the velocity increment at x_i and $I_t(x_i)$ is the scalar brightness residual. A damped linearized optical-flow subproblem can be written

$$\min_v \frac{\mu}{2} \sum_{i=1}^n (q_i^\top v_i + I_t(x_i))^2 + \frac{\nu}{2} \|v\|_2^2 + \frac{1}{2} v^\top G v, \tag{15}$$

where $\mu > 0$ weights data fidelity, $\nu \geq 0$ is an optional damping weight, and $G \succeq 0$ is the regularization Hessian. Setting $\nu = 0$ gives the undamped model. Splitting a copy $w = v$ places (15) in (1).

6.2 Rank-one data Hessian

The data Hessian is

$$H = \operatorname{diag}(H_1, \ldots, H_n), \qquad H_i = \nu I_d + \mu q_i q_i^\top. \tag{16}$$

If $q_i \neq 0$, then H_i has eigenvalue ν with multiplicity $d-1$ and eigenvalue $\nu + \mu \|q_i\|^2$ along q_i. If $q_i = 0$, all d eigenvalues equal ν. Therefore the attained global curvature endpoints are

$$h_- = \nu, \qquad h_+ = \nu + \mu \max_i \|q_i\|^2. \tag{17}$$

6.3 Sobolev regularizer and symbol

The componentwise regularizer has

$$G = G_0 \otimes I_d.$$

For a first-order periodic Sobolev discretization with stiffness $\beta > 0$ and zeroth-order weight $\gamma \geq 0$,

$$g(\xi) = \gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2\left(\frac{\xi_\ell}{2}\right). \tag{18}$$

Thus $g_- = \gamma$. When every Nyquist-frequency corner is present, as on the even periodic grids used here, $g_+ = \gamma + 4\beta d/(\Delta x)^2$; otherwise the exact maximum over the discrete frequency set is used. Higher-order powers use the corresponding powered endpoints.

6.4 Registration-specific four-corner predictor

For each observed local block H_i, consider the globally coefficient-frozen surrogate

$$\widetilde{H}^{(i)} = I_n \otimes H_i, \qquad \widetilde{G} = G_0 \otimes I_d. \tag{19}$$

Each surrogate uses the same H_i at every spatial location and therefore has the exact tensor-product modal decomposition of Section 5. The deployed selector chooses one minimax pair over the family indexed by $i = 1, \ldots, n$; the assembled data Hessian $H = \operatorname{diag}(H_1, \ldots, H_n)$ remains spatially varying and is not replaced in the registration solve.

By (16), every frozen member has component curvatures ν and $\nu + \mu \|q_i\|^2$. Across the family, the attained extrema are exactly (17). Combining them with the attained regularizer endpoints leaves four branches:

$$(h_-, g_-), \quad (h_-, g_+), \quad (h_+, g_-), \quad (h_+, g_+).$$

Define $R_{4\mathrm{C}} := U_{4\mathrm{C}}$ and

$$(\rho_{4\mathrm{C}}, \alpha_{4\mathrm{C}}) \in \operatorname*{arg\,min}_{\rho \in \mathcal{I},\ 0<\alpha\leq 2} R_{4\mathrm{C}}(\rho, \alpha).$$

Corollary 6.1 (Rank-one frozen-registration reduction). *Assume $d \geq 2$, use the globally frozen family (19) indexed by all observed blocks H_i, and suppose the regularizer spectrum attains g_- and g_+. Assume also $\nu + \gamma > 0$, so that $(h_-, g_-) \neq (0, 0)$ and every frozen member is free of a common null mode. Then the worst contraction factor over this family equals $R_{4\mathrm{C}}$, and Theorem 4.3 returns a global minimizer of the frozen-family objective over $\mathcal{I} \times (0, 2]$. For the assembled variable-coefficient operator, the same pair is a predictor rather than a generally optimal pair.*

Proof. For each i, the component curvatures of H_i are ν and $\nu + \mu \|q_i\|^2$, with the former attained because $d \geq 2$. Across the frozen family, their minimum and maximum are therefore h_- and h_+ in (17), and both are attained. Every family mode lies in the rectangle $[h_-, h_+] \times [g_-, g_+]$, so coordinatewise monotonicity places the extreme values of θ at its corners. All four corners are attained across the family, and convexity of $t \mapsto |1 - \alpha + \alpha t|$ therefore makes their largest magnitude the exact family-wise maximum. Finally, $\nu + \gamma > 0$ excludes their common null mode, and Theorem 4.3 gives the stated global minimizer. □

The four-corner reduction is exact for the separable commuting surrogate; the resulting parameter pair is therefore globally optimal over $\mathcal{I} \times (0, 2]$ for that surrogate, but serves as a predictor of the optimal parameters for the full variable-coefficient registration iteration.

6.5 One-shot deployment policy

Algorithm 1: One-Shot Four-Corner oADMM Tuning *Inputs:* initial image gradients $\{q_i\}$, weights μ, ν, regularizer parameters and grid, and admissible penalty interval $\mathcal{I}$. *Output:* $(\rho_{4\mathrm{C}}, \alpha_{4\mathrm{C}})$.

1. From the initial image gradient, compute $h_- = \nu$ and $h_+ = \nu + \mu \max_i \|q_i\|^2$.
2. Compute g_- and g_+ analytically from the regularizer and grid.
3. Form the four endpoint pairs and deduplicate their rational branches $\theta_c(\rho)$.
4. For every remaining branch pair, form the intersection, relaxation- transition, and stationary polynomials in Theorem 4.3. Retain their positive real roots in $\mathcal{I}$, merge repeated roots, and add the endpoints of $\mathcal{I}$.
5. At each candidate compute $a(\rho)$, $b(\rho)$, and $\alpha^\star(\rho)$ (or an admissible default when $b(\rho) = 1$); select the pair minimizing the original objective $R_{4\mathrm{C}}$. Candidates generated by inactive branch pairs or denominator clearing require no special filtering.
6. Run inner oADMM with this fixed pair.
7. In one-shot deployment, reuse the pair at later outer registration iterations and pyramid levels unless an explicit retuning policy is enabled.

An *inner ADMM iteration* solves one quadratic problem with fixed H, G, b, so its optimal fixed parameters do not change during that solve. An *outer registration iteration* updates the deformation and relinearizes; the warped image gradients and H may then change. The one-shot policy reuses the initial pair after relinearization, thereby avoiding repeated parameter selection. The reported ablation tests per-pyramid-level retuning, not separate retuning at every outer Gauss–Newton iteration.

7 Experimental Protocol

The reference two-dimensional solver uses a three-level Gaussian pyramid, spatial image gradients, Gauss–Newton optical-flow linearization, Sherman–Morrison solves for the rank-one data blocks, and fast Fourier transform (FFT) solves for the Sobolev regularizer. Velocity increments are composed with the displacement. A backtracking line search requires objective decrease and a discrete Jacobian above 0.05. Pyramid transfer is audited and, if needed, its nontranslational component is halved until the same floor holds. All methods in a comparison share initialization, objective, stopping tolerances, line search, pyramid, and topology safeguards.

We consider three fixed parameter choices in the comparisons: Fixed ADMM $(1, 1)$, Fixed oADMM $(1, 1.8)$, and the validation-selected fixed comparator $(0.1, 1.0)$. The last pair was selected from $\rho \in \{0.03, 0.1, 0.3, 1, 3\}$ and $\alpha \in \{1, 1.8\}$ on two known-deformation cases from each of two separate public images. Among pairs within 1% of the lowest median mean squared error, the procedure minimized median inner iterations, with median error as a tie-breaker. Neither CIMA nor FIRE was used for this selection. Adaptive comparators are residual balancing and a safeguarded Barzilai–Borwein (BB) proxy; implementation details are given in Supplementary Section 10.2.

The spectral tests use direct eigendecomposition only on tractable systems. The controlled spectral studies use $\mathcal{I} = [10^{-2}, 10^{2}]$; the real-image selector uses $\mathcal{I} = [10^{-4}, 10^{3}]$. Implementations use $\alpha \in [10^{-6}, 2]$ as a numerical realization of $0 < \alpha \leq 2$. The constant-coefficient test uses periodic $n \times n$ grids, $(h_-, h_+) = (0.2, 2.3)$, and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. Controlled noncommuting tests separate periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and random-gradient stress tests on ten 3×3 periodic systems. Direct eigendecomposition gives the exact spectral radius at this tractable size, and numerical minimization of that radius supplies an oracle parameter pair. The precise fields, amplitudes, smoothing scales, and common operator parameters are specified in Supplementary Section 9.2.

For each controlled case, define the normalized commutator

$$\chi = \frac{\|HG - GH\|_2}{\|H\|_2 \, \|G\|_2}.$$

For these tests, let R_{act} be the true spectral radius at the selected parameters, R_{ora} the radius at numerically oracle-optimized parameters, and U a rigorous enclosure evaluated at the selected pair. We report the spectral-radius regret

$$\mathcal{G}_{\mathrm{ora}} := \frac{R_{\mathrm{act}} - R_{\mathrm{ora}}}{R_{\mathrm{ora}}}$$

and the selected-point surrogate discrepancy

$$\mathcal{D}_{\mathrm{4C}} := \frac{|R_{\mathrm{4C}}(\rho_{\mathrm{4C}}, \alpha_{\mathrm{4C}}) - R_{\mathrm{act}}|}{R_{\mathrm{act}}}.$$

The first quantity measures contraction lost relative to the numerical oracle; the second compares the surrogate and true spectral radius at the single pair $(\rho_{\mathrm{4C}}, \alpha_{\mathrm{4C}})$. It does not measure agreement of the two objectives over the parameter domain. Certificate experiments additionally use the enclosure gap $(U - R_{\mathrm{act}})/R_{\mathrm{act}}$.

The real-image pilot uses a public-domain retinal image and an unrestricted immunohistochemistry image under held-out smooth known deformations. CIMA uses the five differently stained sections of the public lung-lesion-3 sample and all ten unordered pairs [2], resized to 256^2 after a common structural-channel preprocessing. FIRE uses all 134 landmarked pairs (14 A, 49 P, and 71 S) at 256^2 [8]. Landmarks are used only for evaluation. Timings use one central processing unit (CPU) thread for Basic Linear Algebra Subprograms (BLAS) and OpenMP. Reproducibility details are in Supplementary Section 12; baseline safeguards and certificate protocols are in Supplementary Sections 10 and 9, respectively.

8 Results

8.1 Exact four-corner validation

The first experiment verifies Theorem 4.2 under its exact assumptions. We construct constant-coefficient periodic quadratic systems on small $n \times n$ grids with prescribed data-curvature endpoints $(h_-, h_+) = (0.2, 2.3)$ and a first-order Sobolev regularizer with $(\beta, \gamma) = (0.2, 0.1)$. The operators commute, and their attained joint modal spectrum has the required Cartesian-product structure. These synthetic systems contain no registration images or landmarks.

The finite algebraic selector returned $(\rho, \alpha) = (0.543512, 1.898152)$ on every grid. Table 1 compares the four-corner value $U_{4\mathrm{C}}$ with the spectral radius of the full reduced matrix $E_{\rho,\alpha}$, obtained independently by direct eigendecomposition. The two agree to machine precision, verifying the predicted equality in this controlled setting. Figure 1(a) displays the same comparison.

Table 1: Exact four-corner verification on synthetic periodic systems.

n	Direct radius $\varrho(E)$	$\lvert U_{4\mathrm{C}} - \varrho(E)\rvert$
4	0.35313044	$< 10^{-15}$
6	0.35313044	$< 10^{-15}$
8	0.35313044	$< 10^{-15}$

8.2 Transfer under controlled noncommutativity

We next quantify predictor transfer after deliberately violating the theorem assumptions. Ten 3×3 periodic systems use spatially varying optical-flow Hessians with magnitude variation, smooth orientation variation, an abrupt orientation interface, or random-gradient stress tests. The resulting H and G do not commute. At this tractable size, direct eigendecomposition gives the true spectral radius, and numerical minimization of that radius supplies oracle parameters.

Figure 1(b) relates predictor quality to noncommutativity. The magnitude-only and orientation-only cases had $\chi = 0.012$–0.139 and spectral-radius regret 0.003–0.087. The abrupt interface had $(\chi, \mathcal{G}_{\mathrm{ora}}) = (0.166, 0.193)$. The random-gradient stress tests varied magnitude and orientation simultaneously, produced the largest commutators ($\chi = 0.171$–0.239), and had regrets 0.238–0.296. Across this controlled suite, larger normalized commutators were generally associated with larger oracle regret; the descriptive Pearson correlation was 0.87. The ten deterministic systems are designed diagnostics rather than a statistical sample and therefore do not establish a universal relationship.

As a separate pointwise diagnostic, the median selected-point discrepancy $\mathcal{D}_{4C}$ was 1.4×10^{-15} and its maximum was 2.8×10^{-3}; this quantity is not plotted in Fig. 1(b). The diagnostic evaluates agreement between the surrogate and the true spectral radius at the single selected pair (ρ_{4C}, α_{4C}). Small $\mathcal{D}_{4C}$ does not imply that the parameter-dependent objectives coincide, so it is compatible with nonzero oracle regret: the numerical oracle can attain a smaller true radius at another pair even when the two radii agree closely at the four-corner point. Detailed noncommuting certificates and scaling studies are reported in Supplementary Section 9.

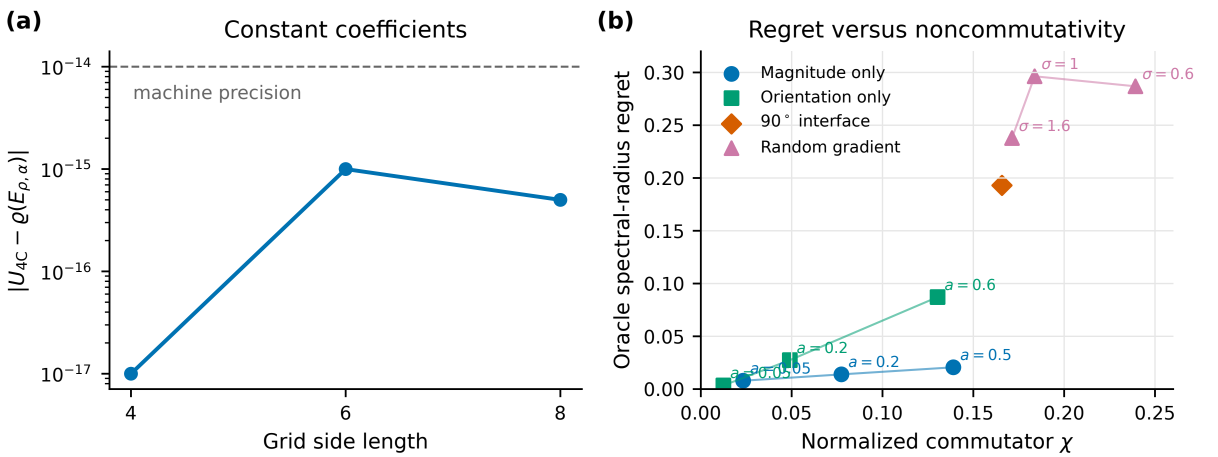

Figure 1: Validation of the four-corner reduction and predictor. **(a)** Constant-coefficient periodic systems satisfy the attained Cartesian-product assumptions; direct eigendecomposition of the full reduced matrix agrees with U_{4C} to machine precision. **(b)** Spatially varying optical-flow Hessians violate commutativity. The horizontal axis is the normalized commutator $\chi = \|HG - GH\|_2 / (\|H\|_2 \|G\|_2)$, and the vertical axis is the spectral-radius regret $(R_{\mathrm{act}} - R_{\mathrm{ora}})/R_{\mathrm{ora}}$. Marker shapes identify magnitude-only, orientation-only, abrupt-interface, and random-gradient constructions; labels give the sinusoidal amplitude a or Gaussian precursor scale σ. Lines connect levels within a construction only as visual guides.

8.3 Real registration benchmarks

8.3.1 CIMA histology

The public lung-lesion-3 sample from the CIMA landmark repository [2] comprises five differently stained sections and all ten unordered pairs. Images supplied at 5% scale were converted to a common structural channel and resized to 256^2; landmarks supplied at 50% scale were rescaled for evaluation only. The problem is diffeomorphic registration of these pairs under the shared pyramid solver, comparing one-shot four-corner oADMM tuning with fixed-parameter, residual-balancing, and BB proxy baselines on target registration error (TRE), iterations, runtime, and discrete Jacobians. TRE is the Euclidean landmark error after registration.

Against the validation-selected fixed comparator $(\rho, \alpha) = (0.1, 1.0)$, held fixed from the independent selection described in Section 7, five order-alternated repetitions of all ten pairs (50 paired runs) showed that four-corner tuning reduced median total runtime by 20.1% (bootstrap 95% interval 19.4–20.7%), solver time by 23.5%, and iterations by 30.1%. Selection used 1.16% of proposed runtime. Median TRE was 7.33 pixels (2.02% of the image diagonal) and differed by less than 3×10^{-5} pixels from the validation-selected fixed comparator. The median landmark-error improvement relative to the unregistered pairs was 36.5%. No global or interior nonpositive Jacobian was observed. One difficult stain pair retained TRE above 22 pixels, identifying initialization and cross-stain representation rather than inner ADMM convergence as the main failure mode.

Table 2: CIMA histology registration at 256^2: medians of iterations and TRE over ten pairs, and of total time over 50 paired runs.

Method	Iterations	Time (s)	TRE (px)
Fixed ADMM $(1, 1)$	1910.5	12.09	7.3310
Fixed oADMM $(1, 1.8)$	1129.5	7.37	7.3308
Residual balancing	1124.5	7.40	7.3310
BB proxy	435.5	3.44	7.3306
Validation-selected fixed comparator $(0.1, 1.0)$	303.5	2.53	7.3306
Four-corner one-shot	**210.0**	**2.02**	7.3306

8.3.2 FIRE retinal registration

For the public FIRE fundus registration benchmark [8], we use all 134 landmarked pairs (14 A, 49 P, and 71 S), converted to a foreground-percentile-normalized green channel and resized to 256^2. Landmarks are used only for evaluation. The problem is retinal diffeomorphic registration under identical phase-correlation initialization, pyramid, objective, and stopping criteria, with one CPU BLAS/OpenMP thread. One-shot four-corner tuning is compared with the validation-selected fixed comparator, residual balancing, a safeguarded BB proxy, and Fixed oADMM on iterations, runtime, TRE, and topology.

Across all pairs, median iterations were 245 for one-shot four-corner tuning and 412 for the validation-selected fixed comparator, a paired median reduction of 39.6%. Median total times were 4.774 and 6.680 seconds, a 34.4% paired reduction (bootstrap 95% interval 30.7–35.7%); tuning accounted for 1.2% of proposed total time. The safeguarded BB proxy required 623.5 iterations and 9.435 seconds; the predictor was faster on 94.0% of pairs, with a median paired reduction of 51.7%. Median TRE differed from the fixed pair by -1.4×10^{-5} pixels, the maximum absolute difference was 3.9×10^{-4} pixels, and no method produced a nonpositive Jacobian. The selected α_{4C} had median 1.5699 and range 1.4483–1.6883, so the upper bound $\alpha = 2$ was not attained. Selected ρ_{4C} ranged from 0.0688 to 0.1402; neither endpoint of $\mathcal{I} = [10^{-4}, 10^{3}]$ was reached.

Table 3: FIRE retinal registration: 134 pairs at 256^2 under identical initialization and solver protocol.

Method	Iter.	Time (s)	TRE (px)	TRE/diag. (%)
Validation-selected fixed comparator $(0.1, 1.0)$	412.0	6.680	5.50178	1.51967
Residual balancing	1434.0	22.325	5.50211	1.51976
BB proxy	623.5	9.435	5.50175	1.51966
Fixed oADMM $(1, 1.8)$	1529.5	23.471	5.50194	1.51971
Four-corner one-shot	**245.0**	**4.774**	5.50177	1.51966

Pairwise percentage reductions need not equal percentage differences between separately reported method medians.

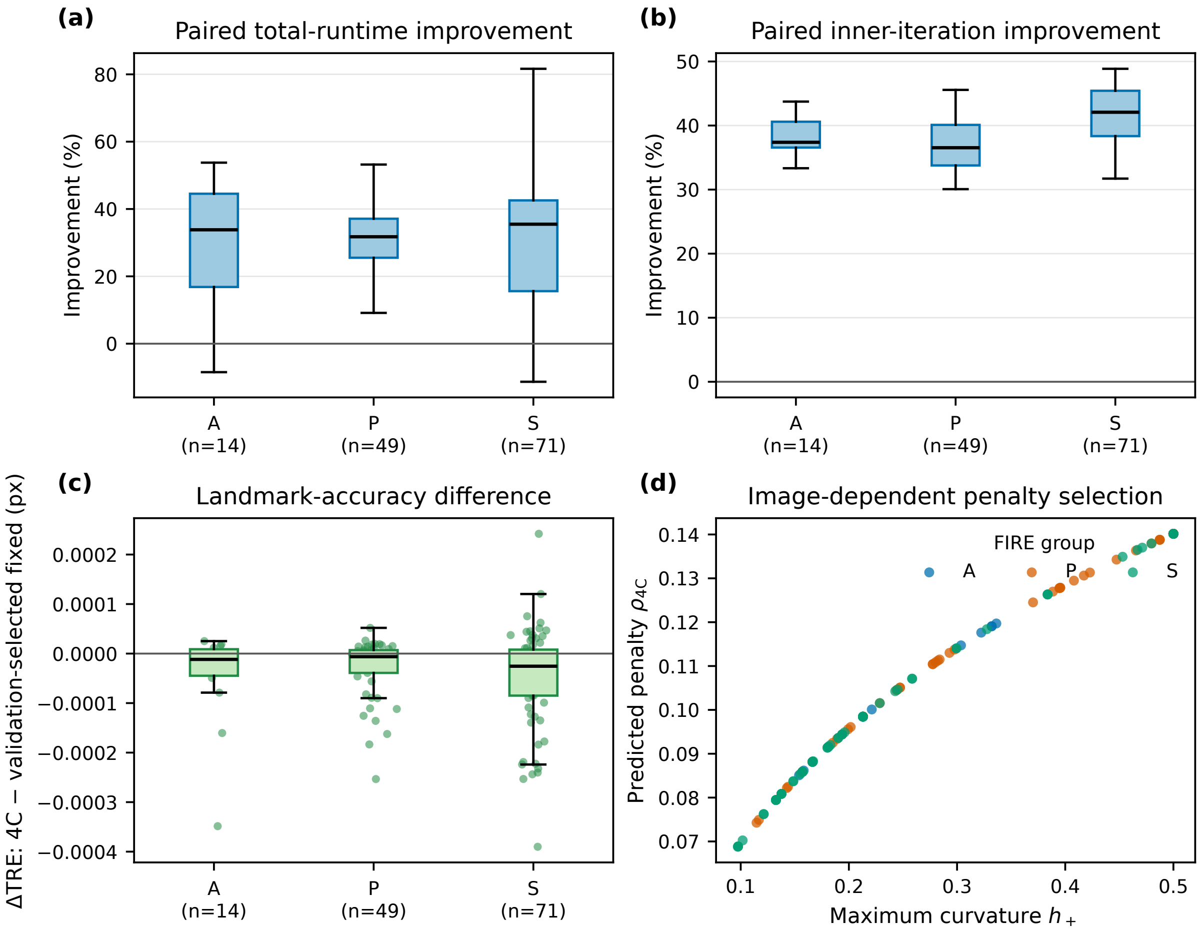

Figure 2: FIRE registration at 256×256. **(a)** Pairwise total-runtime reduction and **(b)** inner-iteration reduction relative to the validation-selected fixed comparator. **(c)** $\Delta\text{TRE} = \text{TRE}_{4\text{C}} - \text{TRE}_{\text{fixed}}$, where "fixed" denotes that comparator. **(d)** Predicted penalty versus image-dependent maximum data curvature.

8.4 One-shot reuse ablation

On a balanced 18-pair FIRE subset, we compared one-shot reuse of the initial full-resolution parameter pair with retuning at every pyramid level. Retuning reduced median inner iterations from 243 to 222 but increased median total time from 2.651 to 3.895 seconds; median parameter-selection time increased from 0.030 to 0.110 seconds. Under this implementation, per-level retuning therefore traded fewer inner iterations for higher end-to-end runtime. The reported method uses one-shot reuse; retuning after every outer Gauss–Newton linearization was not evaluated.

9 Discussion

9.1 Why four curvature endpoints can suffice

Separability supplies independent data and regularizer modal coordinates. Monotonicity along each coordinate line then places the extreme modal values at endpoint pairs, and convexity of the relaxed magnitude preserves endpoint maximization. For registration, the rank-one data Hessian further reduces all data curvatures to ν and $\nu + \mu \max_i \|q_i\|^2$. The resulting four numbers encode the scale balance between the data and regularization terms without reference to image dimension.

9.2 Transfer beyond the separable model

In the assembled registration operator, spatially varying data blocks interact with neighboring regularizer couplings. Smooth coefficient variation can retain locally meaningful curvature scales even though no global common modal basis exists. This provides a plausible explanation for the observed transfer of the surrogate-optimal pair, but not an exact characterization of R_{true}. The predictor omits global orientation interactions among the rank-one blocks.

9.3 Expected degradation and computational tradeoff

Smooth magnitude and orientation variation produced relatively small regret; the abrupt orientation interface and the random-gradient stress tests produced larger regrets; the most strongly noncommuting cases had the largest regrets in the controlled suite. These higher regrets are consistent with increasing mismatch between the frozen separable surrogate and the assembled noncommuting operator. The commutator norm alone does not fully determine predictor quality. More accurate parameter selection can instead use repeated spectral calculations on the large variable-coefficient operator. The four-corner predictor occupies the complementary regime: it uses stronger structural assumptions to make tuning overhead negligible relative to the solve.

9.4 Limitations

The exact theorem requires an attained Cartesian-product joint modal spectrum; registration uses a noncommuting approximation. The experiments are two-dimensional and cover FIRE at 256^2 and one five-image CIMA tissue set at reduced resolution. The optional certificates are conservative, and the tested BB proxy represents one safeguarded adaptive implementation. Finally, the analysis concerns convex linearized subproblems and does not establish global convergence of the outer nonlinear registration method.

10 Conclusion

Quadratic oADMM with an attained Cartesian-product joint modal spectrum admits an exact four-corner characterization of its worst modal contraction. Analytic relaxation and a finite algebraic penalty candidate set then replace iterative spectral parameter search. For variable-coefficient registration, local coefficient blocks and the rank-one optical-flow Hessian define a family of globally frozen separable surrogates, turning this result into an inexpensive predictor. FIRE and CIMA experiments show reduced iterations and runtime with essentially unchanged landmark accuracy and observed topology. Further work may address tighter noncommuting analysis, volumetric registration, and related splitting methods.

Declarations

- **Funding.** No specific funding was received for this work.

- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable; the study uses only publicly available, de-identified image datasets.

- **Consent for publication.** Not applicable.
- **Data availability.** FIRE [8] and CIMA [2] are publicly available from their original providers. The FIRE archive is not redistributed.
- **Materials availability.** Not applicable.
- **Code availability.** Source code, tests, configurations, processed JSON and CSV outputs, and preparation and evaluation scripts are available at `https://github.com/xhxuciedu/admm`.
- **Author contributions.** Katherine Xie and Jie Wu conducted the research and wrote and edited the manuscript. Xiaohui Xie conceived the work and edited the manuscript.

A Proof of the Reduced Cayley Representation

Let $(v^\star, w^\star, u^\star)$ be a fixed point and write $\widetilde{v}^k = v^k - v^\star$, with analogous notation for w and u. Subtracting the fixed-point equations from (2)–(4) removes the affine term b. With

$$P_H = \rho(H + \rho \mathrm{I})^{-1}, \qquad P_G = \rho(G + \rho \mathrm{I})^{-1},$$

the homogeneous v-update and relaxed variable are

$$\widetilde{v}^{k+1} = P_H(\widetilde{w}^k - \widetilde{u}^k), \qquad \widehat{\widetilde{v}}^{k+1} = [\alpha P_H + (1-\alpha)\mathrm{I}]\widetilde{w}^k - \alpha P_H \widetilde{u}^k.$$

Set

$$z^k = \widehat{\widetilde{v}}^{k+1} + \widetilde{u}^k = [\alpha P_H + (1-\alpha)\mathrm{I}]\widetilde{w}^k + (\mathrm{I} - \alpha P_H)\widetilde{u}^k.$$

The remaining updates are $\widetilde{w}^{k+1} = P_G z^k$ and $\widetilde{u}^{k+1} = (\mathrm{I} - P_G)z^k$. Thus, for $s^k = [(\widetilde{w}^k)^\top, (\widetilde{u}^k)^\top]^\top$,

$$s^{k+1} = AB_\alpha s^k, \qquad A = \begin{bmatrix} P_G \\ \mathrm{I} - P_G \end{bmatrix}, \qquad B_\alpha = \begin{bmatrix} \alpha P_H + (1-\alpha)\mathrm{I} & \mathrm{I} - \alpha P_H \end{bmatrix}.$$

Here $A \in \mathbb{R}^{2N \times N}$ and $B_\alpha \in \mathbb{R}^{N \times 2N}$. Sylvester's determinant identity gives

$$\det(\lambda \mathrm{I}_{2N} - AB_\alpha) = \lambda^N \det(\lambda \mathrm{I}_N - B_\alpha A).$$

Hence the nonzero eigenvalues of AB_α and $B_\alpha A$ coincide, including algebraic multiplicity, and the $2N$-dimensional state map has the N eigenvalues of $B_\alpha A$ together with N additional zeros.

Direct multiplication yields

$$B_\alpha A = (1-\alpha)\mathrm{I} + \alpha\{P_H P_G + (\mathrm{I} - P_H)(\mathrm{I} - P_G)\}.$$

Because $P_H = (\mathrm{I} - C_H)/2$ and $P_G = (\mathrm{I} - C_G)/2$,

$$P_H P_G + (\mathrm{I} - P_H)(\mathrm{I} - P_G) = \tfrac{1}{2}(\mathrm{I} + C_H C_G).$$

Therefore

$$B_\alpha A = \left(1 - \frac{\alpha}{2}\right)\mathrm{I} + \frac{\alpha}{2} C_H C_G = E_{\rho,\alpha},$$

which proves Proposition 3.1.

B Proof Details for the Four-Corner Theorem

Fix $\rho > 0$ and let $\mathcal{R} = [h_-, h_+] \times [g_-, g_+]$. The denominators in (9) are positive. For fixed g, the sign of $\partial\theta/\partial h$ is the sign of $g - \rho$ and is independent of h; for fixed h, the sign of $\partial\theta/\partial g$ is the sign of $h - \rho$ and is independent of g. Consequently, replacing either coordinate of any point in $\mathcal{R}$ by a suitable endpoint cannot decrease θ, and a possibly different endpoint replacement cannot increase it. Applying this argument successively to both coordinates shows

$$\theta_{\min} := \min_{\mathcal{R}} \theta = \min_{c \in \mathrm{corners}(\mathcal{R})} \theta_c, \qquad \theta_{\max} := \max_{\mathcal{R}} \theta = \max_{c \in \mathrm{corners}(\mathcal{R})} \theta_c.$$

Thus every attained modal value $t = \theta(h, g; \rho)$ lies in $[\theta_{\min}, \theta_{\max}]$.

For fixed $\alpha > 0$, the function $\phi(t) = |1 - \alpha + \alpha t|$ is convex. If $t = \lambda\theta_{\min} + (1 - \lambda)\theta_{\max}$ with $0 \le \lambda \le 1$, convexity gives

$$\phi(t) \le \lambda\phi(\theta_{\min}) + (1 - \lambda)\phi(\theta_{\max}) \le \max\{\phi(\theta_{\min}), \phi(\theta_{\max})\}.$$

Both interval endpoints are corner values, so U_{4C} bounds the magnitude of every modal eigenvalue. If H and G are simultaneously diagonalizable, $E_{\rho,\alpha}$ is diagonal in their common modal basis and its spectral radius is the largest modal magnitude. When all four endpoint pairs are attained common modes, whichever corner realizes the displayed maximum is an actual eigenmode, proving equality. Without endpoint attainment, the same argument proves only the rectangular upper envelope.

C Proof of Optimal Conditional Relaxation

Fix ρ and abbreviate $0 \le a \le b \le 1$. The four-corner objective is

$$F(\alpha) = \max\{|1 - \alpha(1 - a)|,\ |1 - \alpha(1 - b)|\}, \qquad 0 < \alpha \le 2.$$

If $b = 1$, the second endpoint factor equals one for every α, while the first has magnitude at most one for $0 < \alpha \le 2$ because $0 \le a \le 1$. Hence $F(\alpha) = 1$ throughout the admissible interval, and every relaxation is optimal.

Suppose now that $b < 1$. The unique unconstrained minimizer is obtained by equioscillation of the ordered endpoint factors:

$$1 - \alpha(1 - a) = -[1 - \alpha(1 - b)],$$

which gives $\alpha_0 = 2/(2 - a - b)$. This includes $a = b < 1$, for which both endpoint factors coincide and vanish at $\alpha_0 = 1/(1 - a)$. The function F is convex and is nonincreasing up to α_0, so imposing $\alpha \le 2$ gives $\alpha^\star = \min\{2, \alpha_0\}$.

If $a + b \le 1$, then $\alpha_0 \le 2$, and substitution gives $F(\alpha^\star) = (b - a)/(2 - a - b)$. If $a + b \ge 1$, then $\alpha_0 \ge 2$ and the constrained optimizer is $\alpha^\star = 2$; because $2b - 1 \ge 1 - 2a$, its value is $F(2) = 2b - 1$. The formulas agree when $a + b = 1$.

D Proof of Finite Algebraic Selection

For a corner $c = (h_c, g_c)$, write

$$\theta_c(\rho) = \frac{N_c(\rho)}{D_c(\rho)}, \qquad N_c(\rho) = \rho^2 + p_c, \quad D_c(\rho) = \rho^2 + s_c\rho + p_c,$$

where $s_c = h_c + g_c$ and $p_c = h_c g_c$. Because $\rho > 0$ and $h_c, g_c \geq 0$, every $D_c(\rho) = (\rho + h_c)(\rho + g_c)$ is strictly positive. Identical rational branches are deduplicated.

For distinct branches i and j, an intersection satisfies

$$N_i D_j - N_j D_i = \rho\big[(s_j - s_i)\rho^2 + p_i s_j - p_j s_i\big] = 0.$$

Thus its cleared polynomial has degree at most three (and, after removal of the inadmissible root $\rho = 0$, degree at most two). A relaxation-regime transition for any candidate pair satisfies

$$N_i D_j + N_j D_i - D_i D_j = 0,$$

a polynomial of degree at most four.

Let $\mathcal{Z}$ contain the endpoints of $\mathcal{I}$ and all roots in $\mathcal{I}$ of nonzero pairwise-intersection polynomials. On each connected component J of $\mathcal{I} \setminus \mathcal{Z}$, the strict ordering of all distinct branches is fixed. Hence there are fixed indices i and j such that $a = \theta_i$ and $b = \theta_j$ throughout J. Partition J further at the roots of $a + b = 1$. On each resulting open interval, the reduced objective is one of

$$F_{ij}^{(1)}(\rho) = \frac{\theta_j - \theta_i}{2 - \theta_i - \theta_j}, \qquad F_j^{(2)}(\rho) = 2\theta_j - 1.$$

The denominator in the first regime is at least one and is therefore positive. When $b = 1$ in the second regime, the objective is the constant one and is handled by the piece endpoints.

After common denominators are cleared, $F_{ij}^{(1)} = P/Q$ with $\deg P \leq 3$ and $\deg Q \leq 4$. Its stationary equation $P'Q - PQ' = 0$ therefore has degree at most six. The stationary equation for $F_j^{(2)}$ has no larger degree, so six is a common upper bound. These degree counts also cover coincident curvature endpoints after identical branches have been removed.

The reduced objective is continuous on the compact interval $\mathcal{I}$, so it attains a global minimum. On the closure of each smooth piece, a nonconstant minimum lies at an endpoint or at an interior stationary point. If the objective is constant on a piece, either closure endpoint represents its value. Repeated roots do not create additional pieces. Identically zero intersection, transition, or stationarity polynomials are discarded; roots outside $\mathcal{I}$, including the algebraic intersection root $\rho = 0$, are removed. Because the construction in Theorem 4.3 includes every branch pair, it contains the transition and stationary roots of each pair that is active on a smooth piece without requiring that activity to be known in advance. Roots generated by inactive pairs and roots made extraneous by denominator clearing merely enlarge the finite set. Evaluating the original objective at every retained candidate therefore proves the theorem.

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